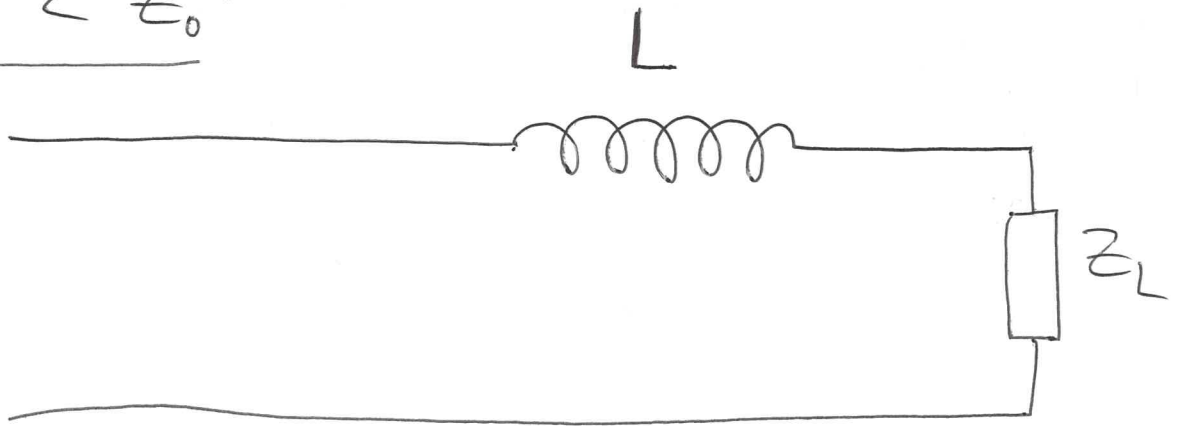
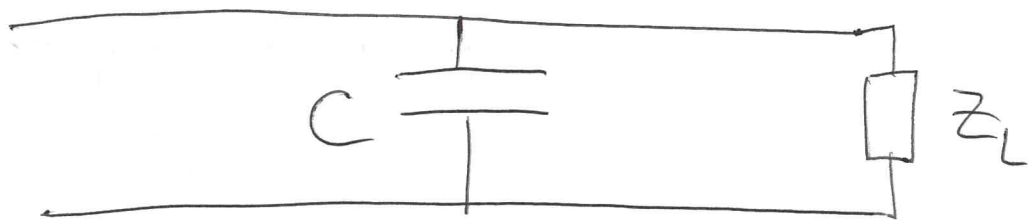


$Z_L < Z_0$



$Z_L > Z_0$

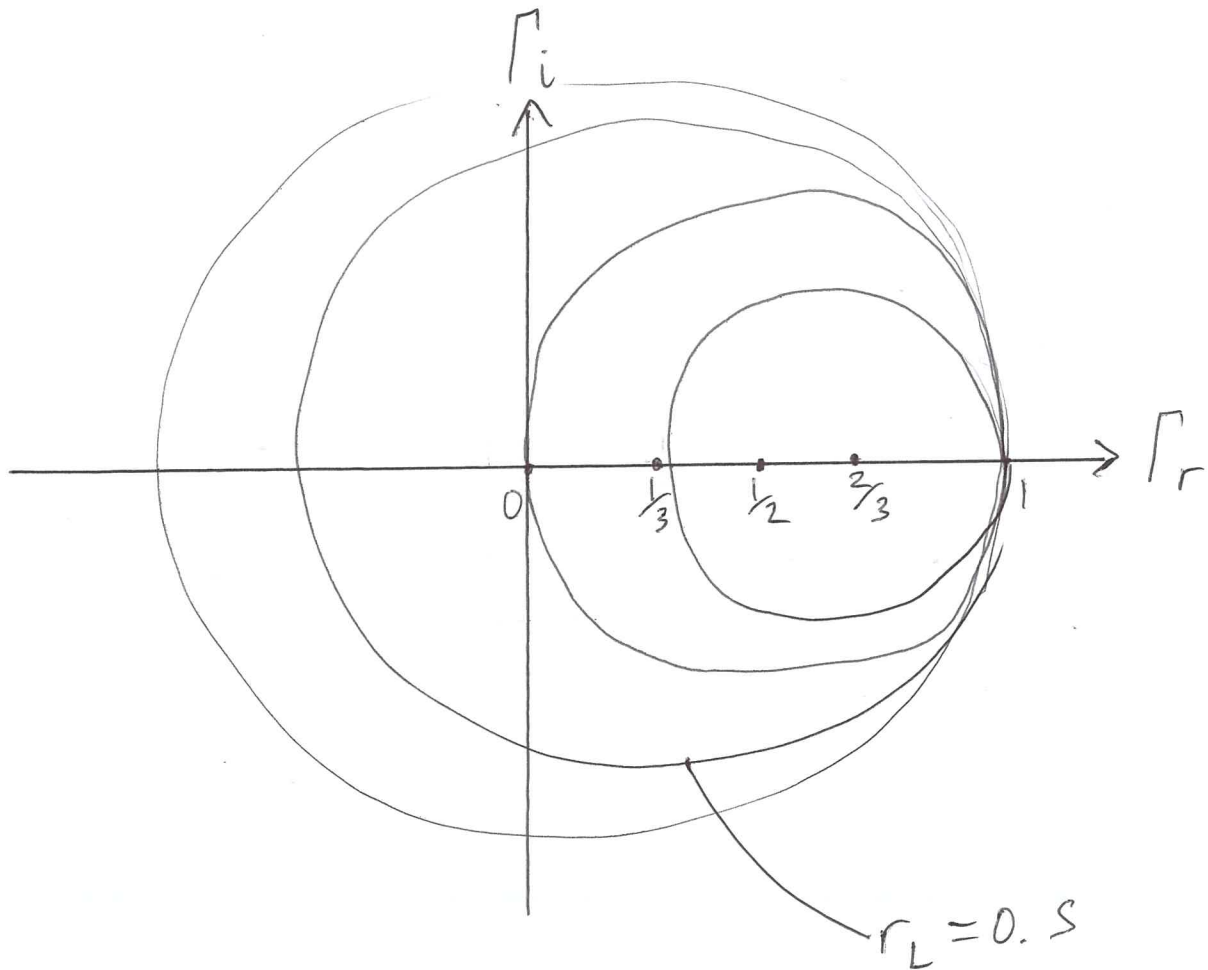


'Lumped elements' approximation

Resistance circles:

$$\left(\Gamma_r - \underbrace{\frac{r_L}{1+r_L}}_{x_0} \right)^2 + \Gamma_i^2 = \left(\underbrace{\frac{1}{1+r_L}}_{\text{radius}} \right)^2$$

$y_0 = 0$



$$r_L = 1, \text{ radius} = \frac{1}{2}, \quad x_0 = \frac{1}{2}$$

$$r_L = 2, \quad \frac{1}{3}, \quad x_0 = \frac{2}{3}$$

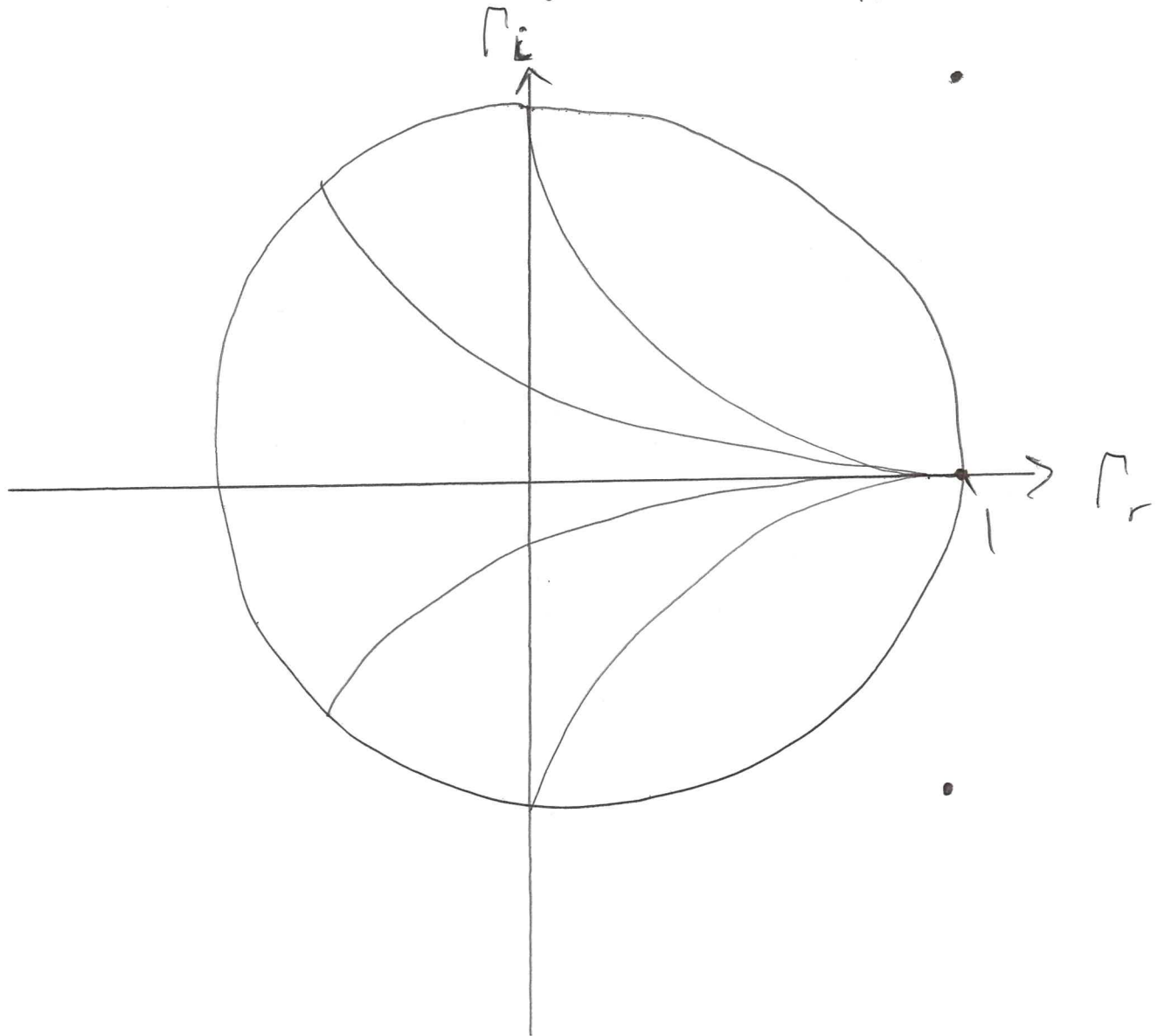
$$r_L = 0, \quad 1, \quad x_0 = 0$$

$$R_L = 25 \Omega \quad r_L = 0.5, \text{ radius} = \frac{2}{3}, \quad x_0 = \frac{1}{3}$$

$$Z_0 = 50 \Omega$$

Reactance circles:

$$\underbrace{(\Gamma_r - 1)^2}_{x_o = 1} + \underbrace{(\Gamma_i - \frac{1}{x_L})^2}_{y_o} = \underbrace{\left(\frac{1}{x_L}\right)^2}_{\text{radius}^2}$$



$$x_L = 1, \text{ radius} = 1, y_o = 1$$

$$x_L = -1, \text{ radius} = 1, y_o = -1$$

$$x_L = 0.5, \text{ radius} = 2, y_o = 2$$

$$x_L = -0.5, \text{ radius} = 2, y_o = -2$$

$$Z_L = 100 + j50 \Omega$$

$$Z_0 = 50 \Omega$$

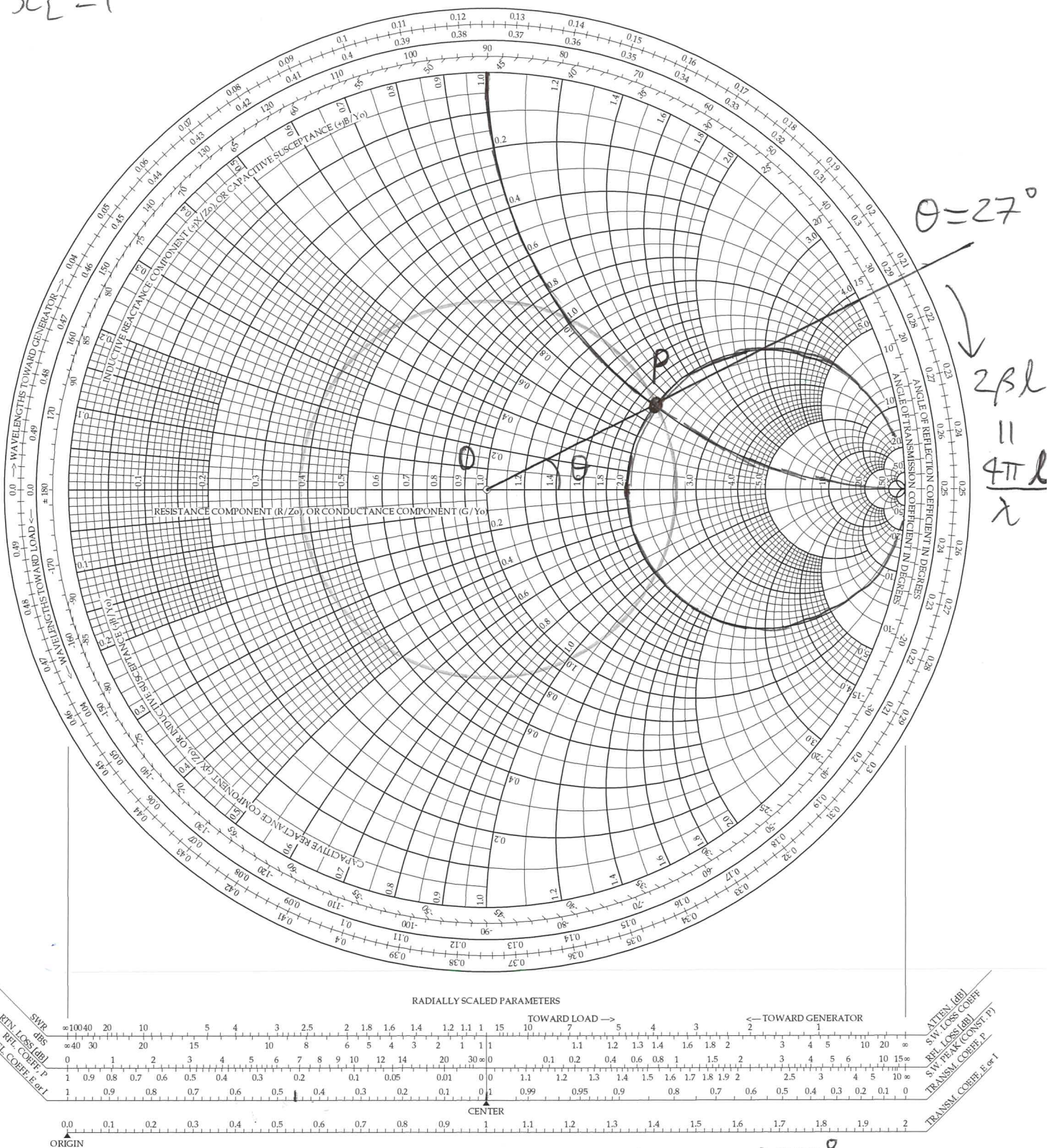
$$Z_L = Z + jB$$

$$r_L = 2$$

$$x_L = 1$$

The Complete Smith Chart

$$\frac{l}{\lambda}$$



$$\theta = 27^\circ$$

$$\frac{11}{4\pi} \frac{l}{\lambda}$$

$$|\Gamma| = 0.46$$

$$\Gamma = 0.46 e^{j27^\circ}$$

$$SWR = \frac{V_{max}}{V_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$RL = -20 \log |\Gamma|$$

(4)

$$Z_L = 3.5 \xrightarrow{180^\circ}$$

$$Y_L = 0.3$$

$$\left(= \frac{1}{Z_L} \right)$$

$$Z_L = 2 + j1.5$$

The Complete Smith Chart

Impedance $\Leftrightarrow$ Admittance

$$Y_L = 0.3 - 0.24j$$

'Flip the chart' by rotating by 180°

